CSCI – 310 Prep. Systems Software / 593 Intr. Operating Systems

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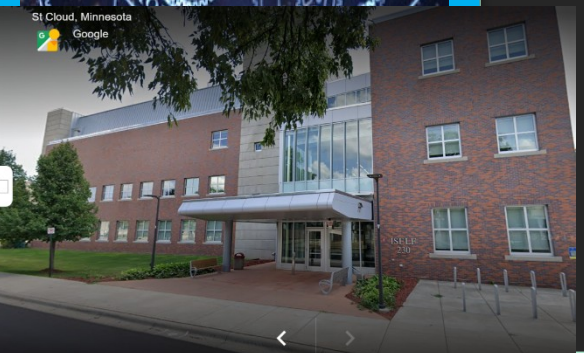

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Office hours:

Check on Syllabus; & by appointment

Note: Use the scripting language you feel mostly comfortable coding.

Avatar Project

Note: optional extra super bounty 2points -- if your implementation works and you decide to contribute with the code to the #3C Avatar Open Source project. The code will be reviewed and voted as the best implementation by the #3C

3C Cloud Computing Club

3C-SCSU

As a club, we get our hands on projects that reflecting new technologies that trendy and in high demand. Reaching us: https://discord.gg/mxbQ7HpPj .

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BOOT CAMP

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TIME MACHINE

FANTASY

FANTASY OUTPUT

CODE FOR FUN

CLASSIC SCI-FI

	1	2	3	4	5
1					
2					
3					
4					
5					

	1	2	3	4	5
1		C		B	F
2					
3		T			
4			C		D
5	M				

STEP 1

		F	M	
			C	
			D	
T	B			C
		F	M	
			C	
	B		D	
T				C
		F	M	
			C	D
	B			
T				C
		F	M	
			C	D
T	B			
				C

OUTPUT

STEP 2

		F		M
			C	D
T	B			C
		F		M
	B		C	D
T				C
		F		M
	B		D(C)	
T				C
		F		M
T	B		D(C)	
				C

OUTPUT

STEP 3

		F	M	
T	B		D(C)	
				C
B		F		
T			M	
			D(C)	
				C
		D(C)	M	
B				
T				
				C

WINNER = Devil

OUTPUT

Create a program with the features:

a) Create four threads that sequentially and respectively represents: Bugs Bunny (B), Taz Devil (D), Tweety (T), and Marvin the Martian (M). Don't allow threads to collide, synchronizing them properly; don't allow. (1 point)

b) Walking is random motion; The Bunny, Taz, Tweety, and Marvin are trying to win the race in their own thread to take control over the new planet x, accessing the worskspace variables located inside the pthread_mutex_lock only, representing their position in the race; print the board every time a player move his position; the competitor has to locate one golden carrot/flag (C) and then run carrying the carrot/flag (C) to the Mountain (F) top; Whenever someone got the carrot/flag (C), show that he is carrying it printing the toon character with the respective symbol: e.g. B(C) ; Whoever place the carrot/flag (C) first on the top of the Mountain (F) will win the race; in the workspace there are only 2 golden carrot/flag (C), which will allow only two runners to try to get to the top of the Mountain (F) first. (2 points)

c) In the attempt to decrease the chances of others to get to the top of the Mountain (F) first, Marvin activates the space x multi-dimensional time travel machine; every time the machine is activated, the Mountain (F) changes location to a random position as any available/empty square; the time machine is activated every time after 3 complete whole cycle. (2 points)

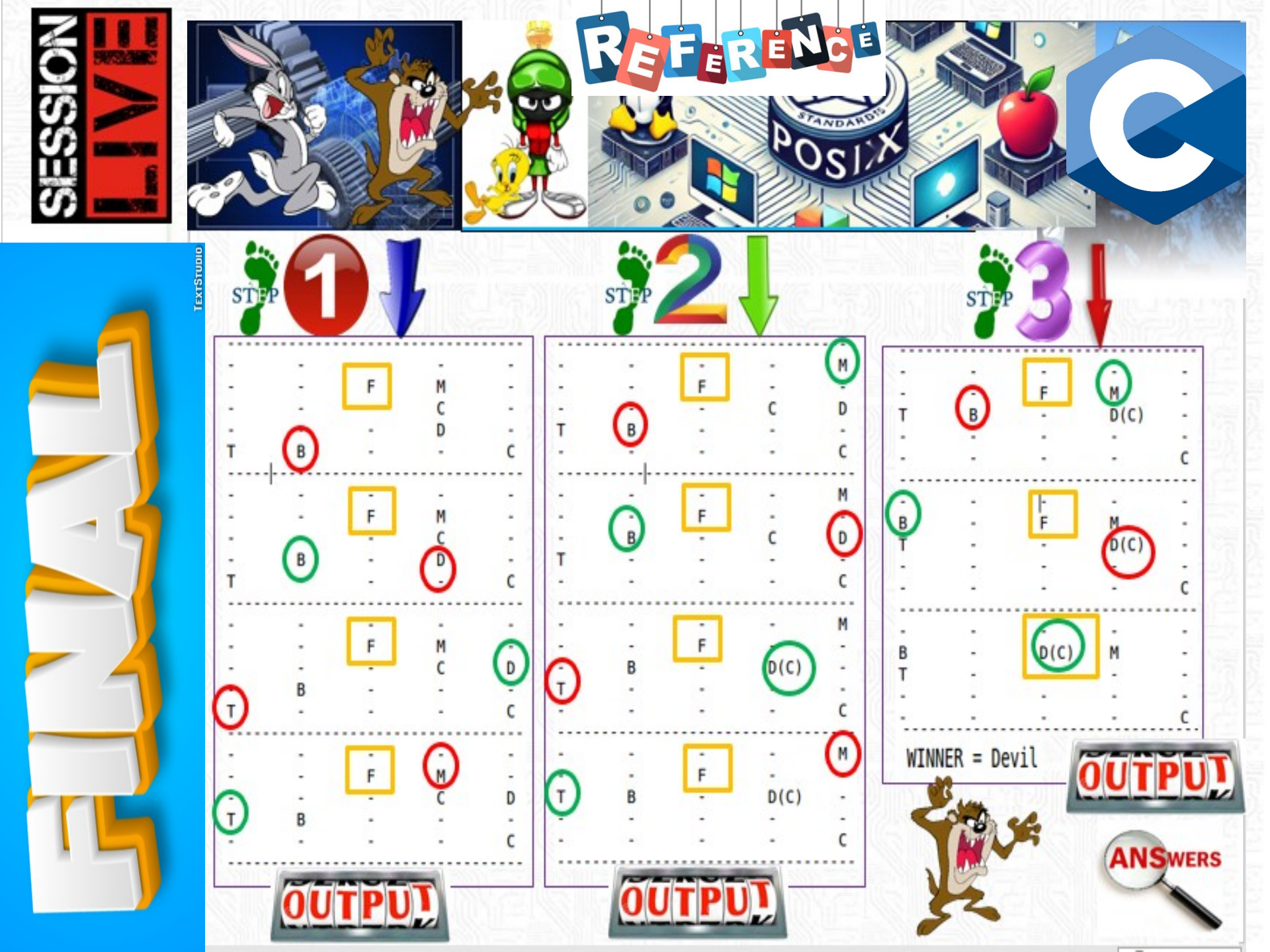
d) Besides trying to find the carrot/flag (C), Marvin cheats, as he is the only runner that is capable of shooting others to eliminate competitors, stealing a carrot/flag (C). If Marvin steps on the same grid where there is another competitor, Marvin will eliminate the competitor shooting, and stealing the carrot/flag (C) if possible; all other competitors are not allowed to step on the same grid/spot from other competitors; nobody is allowed to step on the spot with the symbol mountain top Mountain (F) if not already carrying the golden carrot/flag (C). The race ends when any competitor place the first carrot/flag (C) on the top of Mountain (F). (2 points)

Create a program with the features:

e) Documentation: **describe in word or ppt file the steps taken in your implementation; (1 point)**

f) Submission: **upload on D2L, source code/scripting/setups documentation and a not zipped short video demo 3 to 5 minutes, and ppt presentation; as a comment in the source code, include your name/author in the head of the program. (2 points)**

Note: **The code is to be submitted on the GitHub Avatar Open Source project only per invitation if your code was selected. That will add the bounty extra credit point.**


SESSION LIVE
REFERENCE
POSIX
STANDARDS
FINAL
STEP 1
STEP 2
STEP 3
WINNER = Devil
OUTPUT
OUTPUT
OUTPUT
ANSWERS

GitHub

open source initiative®

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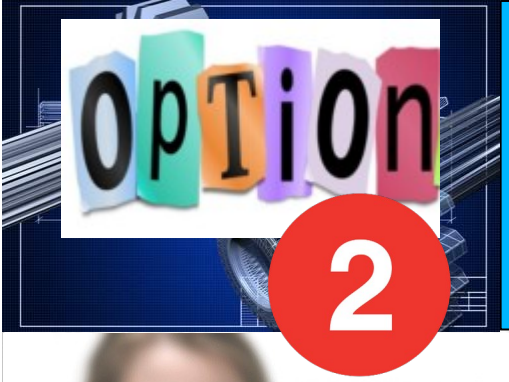
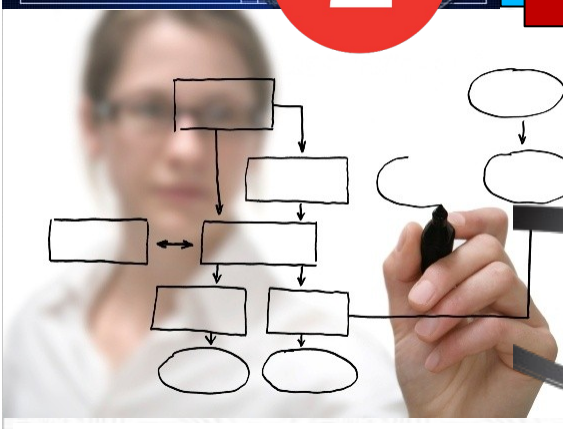


Challenging OPTIONAL PROJECT

Open Source



Practical DEVELOPMENT

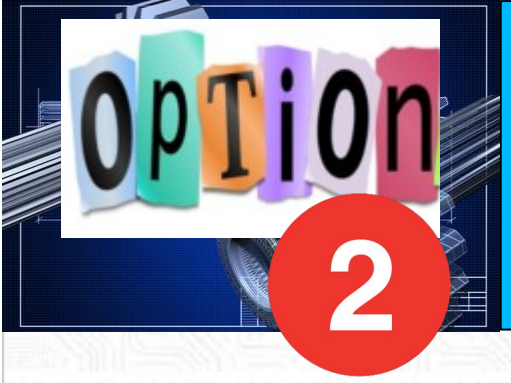
open source

Labels

DevOps | automation | bash | enhancement | file management | python

FIRST COME FIRST SERVED

Note: Team of 3 Implementation



Challenging OPTIONAL PROJECT

Open Source

Open 47 Closed 171 https://github.com/3C-SCSU/Avatar/issues Author Labels Projects

Artificial Intelligence: Fix Tab Outside GUI A.I. design faulty GUI-v5.0 project consistency python Qt Software Quality Assurance usability
#571 · 3C-SCSU opened 15 hours ago

Server: HDD installation benchmark benchmark Big Data configuration hardware server Setup
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